ActInf Livestream #011.2 "Sophisticated Affective Inference Simulating Anticipatory" (2020)

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hello everyone welcome to the active inference live stream it is active inference Liv stream 11.2 on December 29th 2020 this is our last group active inference live stream of the year so thanks so much everyone who is out here today and also um watching live for those who are and I will share my screen for the participants and here we go Welcome to the active inference lab everyone we are an experiment in online team communication learning and practice related to active inference you can find us at our website on Twitter email YouTube or our public keybase team and username this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and as far as video etiquette for live streams mute if there's noise in your background raise your hands so that we can hear from everybody on the Queue and we'll use respectful speech Behavior as I stated this is the last active inference live stream for 2020 but in 2021 it's already shaping up to be a great year we're going to be meeting Tuesdays from 7: to 9:00 a.m. Pacific Time PST um or Pacific time when the time changes and you can go to this link to learn more when you go to that link you'll see this spreadsheet and it has instructions on how you can participate as well as which papers we'll be reading in which weeks so uh you can see that many authors will be joining us and hopefully everybody will have the time to read and digest these papers today in active inference stream 11.2 we are going to have introductions and warm-ups and then we'll go into the 112 itself we're going to be continuing the discussion of sophisticated affective inference the paper of hesp at all 20120 welcome Sasha we'll run through again the aims claims abstract and road map and the figures and Happy New Year's everyone thanks a lot we'll do new stuff in 2021 and so stick along let's get to the intros and warm-ups we will introduce ourself just by giving a short introduction or check-in and then passing to somebody who hasn't spoken yet so I'll start I'm Daniel and I will pass it first to a firsttime visitor Scott hi I'm Scott David I'm at the University of Washington Applied Physics lab and I've been a lawyer for 27 years I'm trying to use physics principles to help structure social relationships and I will pass it to Stephen hello I'm Stephen I'm based in Toronto I'm currently doing a practice-based PhD with Canter Christ Church University and I'm looking at applying the dynamic processes of active inference into my work with participatory theater and Community Development and helping to understand the Dynamics of all of that and I will pass it to Ivan hello my name isan I in Moscow and I'm a researcher in system management School uh pass it to Blue hello I'm blue Knight I am an independent research consultant based out of New Mexico and I will pass it to Sasha hi um I'm Sasha I'm based out of Davis California and um I study developmental neuroscience and I will pass it to uh Alex hello my name is Alex I'm in Moscow Russia I'm also researcher in systems management school and I'm trying to find a ways to connect active inference and systems engineer framework thanks cool and Ryan uh it seems like has the same situation as last week but definitely for 2021 we'll figure it out um I'll just tell them it still does not work all right let's go to the warm-up questions and again if anyone else joins we'll just roll with it so for these warm-up questions raise your hand it'd be great to hear from everybody let's just start off with what are some kinds of agents that we might want to model in active inference so just even before reading this paper or after reading the paper what was a kind of system or a type of agent that we might want to frame in this way of thinking so emotions makes us think about individual people but there's other kinds of systems we might want to model as well um actually Scott you were mentioning some interesting uh types of Agents we might be considering right before we started so maybe could you pick up there sure um I was mentioning in law it feels like one of the things one of the strategies is to introduce a provisional agent a nominal party into an interaction so if you have two people interacting you can introduce a third by introducing the third it changes the system because it's now a three-party system with three-party concerns and that system then will have a different markof blanket as I am beginning to understand it and so in that context you may be able to do things you can't do in the other through the other markco blanket because it may be able to an example is in in tax there's a thing called a like kind exchange where I can exchange commercial property for another commercial property taxfree well very often you can't find someone with exactly the property you're looking for let's say you have a 20 Acre Farm you want a 30 Acre Farm there's nobody around with a 30 Acre Farm you use an exchange facilitator and that means there a third party in there they take possession possession of both properties and act like a mixing bowl and so putting the details of that aside it essentially changes that system so changes your Markov blankets and so one of the things I'm wondering about is the intentional shifting of the Markov blankets to increase solution space cool so we have all kinds of agents who are going to be interacting and potentially by Shifting the boundaries the interfaces between agents and wrapping them or subpartitioning them it's possible to make new kinds of swaps um within existing infrastructures or new ones so anyone can just raise their hand I'll just put up the second question which is another General one which is how can we learn and communicate the technical aspects of active inference in an accessible fashion Stephen go ahead I don't know if I'm going to explain how to do this but I'll explain a challenge which um um but I think it's important to overcome it because it's so valuable is I was just in a conversation and trying to talk about systems and how we can how we think about systems when we engage um in communities and thinking about how we understand the world and there's a lot of stuff in in in the world of kind of the social arts and participatory theater where it's like we need to do something with the system and then it flips to we're all a living system and it's the body knows and we just know how it is we just need to listen to the body and it will know and um it's both you know it's we're in this Dynamic niche of which there's an awful lot of stuff we don't know yet we can tap into our body and the way that our body works and there's a way that we construct meaning in the world so I suppose trying to find a way to take case studies and show how it can help would be really useful um thank you Sasha yeah um thanks Stephen I really um agree with what you said and uh ironically or I guess in a very meta way um Act of inference is the answer to that question because um through like closing the loop and introducing action into the system we can better understand thinking about the system and so I I think that's an really great point that um case studies and uh really specific applications of active inference are going to be the best way to teach and learn about it um so yeah I think that's a great Point cool blue so just to kind of piggyback on that I think um and maybe tie in some of what Scott was saying earlier uh when we think about um modeling in active inference it's interesting to think about scaling this not in the way that we talked about when we were talking about scaling active inference but in the way that like we have several different systems even just existing within the body right like the input and output of the stomach and the spleen and the liver and and you know also like the brain and the the sematic awareness right to speak to kind of what what Stephen was saying earlier so it's interesting to try to think about modeling systems within systems in in active inference yep and actually this brain and body uh dialogue and the multi systems approach in the way that it doesn't say well everyone told you the brain was important actually it's the body no it's actually it's both and similarly it's like quantitative and qualitative and so it's not just that we're going to throw out one for the other but we want to think across that Division and integrate them that's the Beyond internalism and externalism and integrating The Sciences with the humanities and other things that we're kind of talking about Scott I just put in the chat there was an article that came out in December 4th in Science magazine called the exploitative segregation of plant roots and so this is helps with that second question uh learn communicating it what it was talking about is the way that plant roots grow specifically you have two trees next to each other each has a system each has a markof blanket they're growing next to each other and they differentiate their plant growth roots to optimize statistically their nutrient uptake and so they don't go and intrude on other plants Roots space because that would not be optimized for them because they'd be putting more resources into into growing further roots and and but getting fewer nutrients because they'd be nearer the other plant and so to me I looked at that and I went wow that's a real nice way to understand the Dynamics of two different organisms interacting with statistical gradients there it's a nutritional gradient specifically that felt to me like it was a nice way to kind of introduce that idea of the of the contact between the two different systems cool so what I'm going to take from this second question and all the an to it is that through examples that are accessible and at hand we can all start to learn more about the relational side of active inference and also the statistical and the quantitative because both those sides are going to be learning Journeys for everybody because no one's going to know all the statistics no one's going to know all the Deep relational pieces that are so uh familiar to Performance artists and other areas the last question on the warmup is what is something that you are wondering about would like to have resolved by today's discussion so one thing I'm wondering about is um was raised right before we started as well is like how do we think about emotion or this parameter in systems that aren't us and I'm also even wondering whether the emotions we experience are these parameters and how do we name parameters or talk about parameters of mathematical models in a way such that they're recognizable and capture meaning but also don't constrain or sort of uh falsely limit what it means to be that term so we can get into a map territory issue when we're talking about computational models of emotion just like we could with Consciousness what you were saying about um emotions like I don't know as I'm starting to study more about emotion like I'm a a logic and science kind of researcher and so but learning about emotion and just reading personally a lot of it seems to be like non-verbal right and so how do you like quantify or even measure emotion on a scale when it's it's really like you know you're you're feeling some you know primitive thing from you know when you were left in your crib as a child that you have no words for because you didn't have words for it at that time and so how do you kind of like like impart some kind of meaning or quantitative scale on something that that maybe there's not even language for yes it's almost like the uh discredited but guiding three-part brain model so it's like the most prefrontal is the numbers in that level of formalization and then a little bit lower we have language but then below that the substrate is actually experience and that's the embodied experience that may or may not even have words to express it and then we can look at expressions like poetry or speech and is there a way to get more precise or is it merely an expression that's arising from an emotional state but the emotional State isn't a number it's not a words it's actually something that's more deep Scott I was wondering if with regard to dance and performance whether there are ways of communicating because we talk about embodied I wonder about encoding so can emotions be encoded in dance in a way that they can't be encoded in speech or other movement so that it becomes a surrogate or kind of that mirror neuron idea that you're projecting out an emotion by the projection of your body position the morphology uh being the encoding of the emotion cool question yeah Stephen definitely yeah I think this also relates to this question of um affect and dimensionality so maybe there's a way that when we take a perspective on things we kind of look at a dimensional approach but it can be encoded in some sort of spatial um aect Ive way and maybe those Dimensions aren't they they they become like Dimensions but the encoding is is not as linear as that a bit like color space when we see colors the the space of colors you can encode it in sort of in in lightness and two different types of color Dimensions but the way that it's encoded is when people look at perception it's not it's not linear in that space so maybe there's like an affect space which somehow encodes in a in a way which is n-dimensional so mate oh sorry go ahead go ahead so might um Aesthetics be um provisional um efforts at encoding very nice to bring in the Aesthetics and the intuition because intuition is that which can't be expressed it's sort of like the dark matter it's the submerged part of Iceberg but those reflect the deepest priors potentially that help us search through the state space that as we've been exploring is vast to do the exponential rollout of a chess game is going to be hard it's going to be more um basically States than you can keep track of however if you think I feel good about this uh arrangement of the pieces then with just a simple affect and a limited roll out you may be able to do very well so again we're not sure but we'll figure this one out okay uh let's is that just one other thing is that like the holographic principle potentially that on a system the information about the system is displayed on its surface is a Markov blanket potentially that surface that you observe to understand the system inside I believe so and Chris Fields who will be coming onto the stream we'll be talking about a paper that's going to be uh in early March Chris has done a lot of holographic marov blanket research so let's keep on thinking about that cuz that's very interesting I'm not exactly sure but yeah hopefully we'll figure that one out within a few months so the paper itself and then again we'll have enough time just to hear whatever people are thinking but uh sophisticated affective inference we talked about it in 11.0 and 11.1 and the paper is about combining these two previous threads of research which were sophisticated active inference which is sophisticated meaning deep through time with affective inference or active inference where there's an emotional state component the abstract they state that and the road map of this relatively short paper is pretty short so today let's think about how these two threads which are distinct threads that's why they each were their own paper can be combined into something that is in front of us today which is this paper that is literally sumting these two threads of research and then we can think about a few directions that this research makes us think about whether developing more advanced simulations so just asking which parameters of the simulation could be improved or how could this be expanded or generalized but also on the more functional side we can look to computational Psychiatry because we're talking about emotions if we are and also even nonhuman agents like robots so few different directions we can go and to structure that discussion it could be useful for us to have a few examples like two or three or four of sophisticated affective inference agents conventional or unconventional so certainly one of them could be a person but we'll think of that person being in a certain context and then we'll return to these examples that we come up with right now as we go through the slides ahead we'll just return it to example a BC and then when we're talking about parameters we'll be able to say okay so in the example a this Matrix would represent this in the example B it would represent that so who has a thought or a suggestion on what would be like a cool system to bring into this mixture so again one of the examples will be a person and people can suggest what they think that person's like story or situation should be kind of like an improv scene and then also if people want to provide uh an unconventional uh system that'd be good too so any thoughts on that Alex um I think maybe we can consider the next system scale and see on team Behavior great and uh so try to Define uh these uh Concepts in a meaning of team communication inside and outside communication with different subsystem of organization thanks for suggesting that so let's do um example a is going to be an individual human so that will be hopefully grounding us to our lived experience and to realistic situations that people and comic book characters find themselves in the second example B is going to be team communication so we'll kind of develop the story but we'll see maybe the team is a startup team or maybe they're a research team or maybe it's uh just an affinity group about some other topic and then the third example let's think about something more organizational and institutional so let's think about that third example being potentially like uh a large company large enough to have a legal department and an HR department let's just say so we have three examples which is going to be the person the team and the institution SL Organization three levels of analysis and then as we walk through these formalisms let's see where might be like strengths weaknesses limitations insights Etc Stephen and I suppose you could apply it to all of those this idea you know we have this vision and Mission I mean that's more for an organization the mission and the vision transfers down to a team which might be the purpose and the roles and then you've got the kind of the character and the kind of you know meaningfulness for the individual so you've got this tempor depth being sort of expressed across all of those levels of scale yeah exactly like the examples in this case they're not three disjoint examples in fact the individual could be on a team in an organization so all three of these examples that we're going to be talking about could be nested within each other or they could be disjoint and so that really highlights why it's so important to have a multiscale perspective because even when we're talking about the individual we're going to be thinking about the individual it's the system of Interest it's what we're talking about it's the interface we're talking about the marov blanket that we're talking about but it's inherently embedded within a larger system so we're not going to forget the company when we're talking about the individual and vice versa so it's always all the scales happening at once but then depending on the system of Interest or the marot blanket or the level of course graining that is being uh performed in that specific analysis it will uh will dive into the details as they present themselves but that will be kind of fun so here's figure one in the paper and we went a little bit in the previous videos into some of these subcomponents because this is like M4 and there's M12 3 as well that lead up to M4 but just to recap what each of these pieces are o at the bottom is observation and S is the state that inference is being done upon so this is where those examples come into play so in the individual case the state that that something might be having uh that an individual might be performing inference on it could be is it sunny or cloudy outside and maybe that changes how you feel or it could be a hidden State like is this person um happy with me or sad with me um a is the mapping of how the hidden States relate to the observations so whether at the individual level it's sunny outside and then the observations are perceived with certain probability um and then yeah people just raise their hand if they want to like jump onto another example or uh you know ask a question about how some of these variables might interact I'll just walk through them with no example and then we'll go to the example pieces B is how the states are inferred to change through time so the B is like the probability that given that at State one it was in such and such a state that at the next time point it would be at a different state now U is a piece that connects to B and this is the policy and so the policy is something that influences how states are likely to transform into each other the policy itself is Downstream of the agents decisions about uh which actions to engage in it turns out that the way that the action selection is chosen at the level of analysis whichever blanket that we're talking about that is uh action selection Guided by this set of parameters the G is that expected free energy that's being minimized and that also covers up the C Matrix which is like the preferences the gamma is the Precision or the uncertainty whether it's like one over or not and basically this relates to how precise the agent believes they are in their estimate of which policies to be pursuing given this chain of things happening down here a um is the uh is the okay sorry about that at each time point in this model because I want to I want to General but I want to be specific about what this model actually did at each time point in this model that's being captured by this top sort of pinkish area this is the affect of inference question where the affect of the agent is uh relating into how it simulates rolling out in this whole internal Gray Zone all this these future time points and um then each model tick is going to be like moving from the left to the right and at each time point in this model the agent is going to be running out some uh possibilities for what could happen in the future so Stephen and then we'll think about the examples yes sorry I was going to sort of bring an example there so maybe go ahead go ahead yeah um well one thing that's come up and I I've been doing some work with this metaphors of movement and Andrew Austin who does that work he talks a lot about how anxiety he finds anxiety is often related to people having trying to have elevated status like they're trying to have an elevated status and it's and it makes a lot of sense with this because it could also be like you feel like you put yourself in an elevated status and you're always going to be slightly uncertain unless maybe you're a king or someone about that status being maintained so being put on a pedestal can be so there could be something about Western society and Status which is partly why we have this overarching anxiety because um and it would make sense in a way because there's always a sense of never being completely clear about whether your status is going to be maintained so I I thought that's kind of interesting um and this would sort of tie would support that kind of observation yep so status or um various other sort of social approval metrics we can think of it's like the water we swim in as social organisms so just like like a company might be wanting to stay above water financially at least that department a person has to be staying above water reputationally we can think of it in that way and so again let's go to the examples and think about these different systems and what they might be performing inference on Scott just going to just going to comment on the status U issue that it feels like then you can adjust so you're making serial adjustments right you're trying to minimize the free energy minimize the difference between your expectation and the reality you're experiencing as I understand it so with that overall effective status you can do what feels like compensating controls so you could if you feel like you're at risk Visa V the one presentation you're making of self you can rake in other presentations of self to complement or supplement or mitigate the impression of one of them because because you have Myriad effective projections that you're making simultaneously so it feels like if you have an awareness of this it can help you with that presentation of self-question by understanding that you okay this effective status that I'm presenting is not getting me where I want to be inside so let me leverage another one which is contiguous or a neighboring affect I'll call it an affectation here but I mean in a present ation of a marov binding site um so it feels like it gives a strategy for hybridizing and synthesizing your presentation of your Markov blankets anyway keep going just it's impressional yes okay so let's go to the examples and think about how these three potentially even overlapping systems are going to be carrying this out and an individual at each time point maybe let's just say this is every second or each minute a a team might be interested in organizing on the multi-day and the organization is going to be more explicitly thinking okay quarter to quarter that's going to be our next time that we're going to do all this simulation so it's not surprising that as things get bigger they also get slower so that's a common theme we're going to see so the individuals affect is really where this emotional inspiration comes from so there there's the closest mapping between affect and emotion but let's think about that team and the large company example well affect again and veilance specifically is like that plus to minus axis so is this team going well is this company going well and if someone says yeah we're burning money at an incredible rate it's unsustainable but it's going well because I have a uh believed trajectory that we're going to go on where we actually pull out of the nose dive and do XYZ so they can still think it's going perfectly well even if the instantaneous rate of loss is high and then on the flip side you can imagine a team or a large company where instantaneously they're state looks like it's good it looks like they have a lot of cash or it looks like they have a lot of people involved in the team but they still feel negative about it so that disjoint between the affect about a situation and the state of the situation is one thing to take into account now let's connect the affect at all these levels to Precision estimates at different levels um especially in the active inference framework which frames what organisms or any system is doing as Precision seeking high anxiety is going to be coming from having low Precision now it's not that we want zero um uncertainty we want to have this controlled novelty sort of U curve but it can be clearly uh situations where your Precision is too low and then you're surprised by what is being uh emitted to you in the niche whether it's your team operating environment or the company's operating environment and then if your Precision it can't get too high because if you're overly precise and you're incorrect then you're going to die like if you have a super super accurate understanding but you're just delusional then you're going to fall off the cliff so in a way what shapes this Precision is actually the real world which prunes beliefs that are not functionally associated with success in the real world Stephen and then we'll come back to it yeah I really like this um this idea of of thinking about how we understand ourselves within all of this so for instance yeah if you could move between a different regime of attention or sense of self which could be like what happens when we do sense making maybe even change the operating metaphors of the person you know who am I at the moment you know what's the context what's my um situational um Attunement what's my team's purpose what's what's the operating metaphor for this team are we like all driving together in a car or we are we building ourselves a fortress you know and that would that almost could give a way to jump in and out of these kind of effective engines if that makes sense so as you seen over time how's it going what's the policy um and what's the the affective States um that sort of thought about in a certain future and they could almost then say okay what if I shifted maybe that's where You' need the body like the metaphor I'm working with and then what does that do and um and maybe at times that's when you need to work with the group to be able to do that because maybe an individual can't just keep flipping backwards and forwards you need to work as a group to do that kind of sense making across different times cool Scott I had a question for Sasha you were mentioning pruning but you weren't talking about dendritic pruning but I wonder if dendritic pruning is describable by what the type of pruning you were just talking about because my understanding of dentritic pruning very minimal is that you know during the day you have the growth of dendrites and then this pruned back selectively um and that that is influenced by your externality each day the growth is influenced by your externality and then there's this pruning process that feels like it's iterating itself at that at that level is that describable by this here is that another scale or is it something else that's going on there um yeah I think that's a really interesting analogy that that's um kind of what's happening there I I don't know if I would uh kind of draw all the same parallels um yeah just from a mechanistic standpoint but it's true that kind of in the uh in the generalities that um uh as connections are formed during the day then um they're pruned or lost um as they're not used and yeah we think that happens at night um but yet I I think that's an interesting um analogy to draw that it's uh beliefs about the world that aren't um I don't know useful or supported are the ones that then get pruned yes oh um I was also going to say uh the other aspect is I think we know uh from our lived experience is that it's really um it it's really interesting when people can be uh holding certain beliefs and not um not know that and then when they finally kind of confront the the belief um it kind of yeah there's a bit of a breakdown and um to me that that moment is very interesting um when uh people have to confront certain beliefs that um maybe they didn't even recognize that they held so there's a phase change and a recontextualization piece in the last part you said Sasha and the pruning is using a network analogy so it's like pruning connections between different aspects of the system and so let's think about the individual in the uh office having a conversation with somebody the query the prompt to them let's just say is how are you feeling and that is a query that's presented to the interface which is their ears and also their vision and other aspects the query is presented to the interface about what's inside of the interface and now what's happening inside of the interface is a generative model of that query presenting system so inside of the person there's a generative model of for example the person who's talking to them asking that question and we know it's a generative model because if they covered up their eyes or if they only heard part of it they might be able to still recover what was being asked but in either case there's a generative model inside of the interface and a prompt is being presented as an observation stimuli to the interface that's querying how the system will respond so speech is action and that's not even uh a legal claim it's that it's a motor claim speech is generated by a motor Behavior it's not your hand but it is a motor behavior and for some people it is with their hand so various body parts can be used to communicate let's think about that interface in the process of the team and the company so potentially for these organizational entities the interface could be like a consultant who interfaces with specific questions but instead of a counselor who asks how are you feeling to an individual there might be questions that could be asked to the team and there can be questions that could be asked such that only the team could respond and someone might say well how could you ask a question that only a team could respond to aren't teams composed of people there's two things I'd say there the first is that that's the reductionist argument the reductionist slide is like what do you mean get a response from a person aren't you just to get a response from skin cells and neurons it's like right that's what I'm trying to get that response so even if the response can be reductionistic understood as being composed of smaller pieces that's not a surprise that should be taken for default so questions can be asked to teams that of course humans are going to be responding uh to as well but also we can think about kind of like Colony phenotypes we can think about team phenotypes so for example let's just say I ask a question to a team like uh what did you have for lunch or what do you think the most exciting project is for next year and then I looked at the distribution of how long it took people to respond to that message that's like a colony level trait it's a collective trait that distribution is composed of people acting but no individual can dictate what that distributional parameter is so it's like a measurement of let's just say some blood cells you flow a million blood cells through and then you can get a distrib ution across all of those blood cells so it's like you're querying in the aggregate even though each individual sub piece uh could have been following that distribution or uh somewhat of an outlier so those are the kinds of things that you could ask presenting to a team or an organization and the team or the organization is going to have a generative model of its Niche so if it's involved in a certain type of consultancy where it paid a lot of money and it really respects what the person is going to be uh doing then maybe when you ask them the question what's the most exciting project and why they give one response or what's the most challenging project of this year and why was it challenging whereas if you were in a different context you'd give a different response because the generative model of the system across that interface is very different so the response that it gives will be different and it's going to be querying different things and externalizing different aspects of itself could be accurate could be inaccurate Scott so one of the things that raises is I've been in some of the work that I've done in cyber security I've been parsing principles versus ethics versus norms and so it's kind of interesting in querying a group you're going to get an average right you'll get a gaussian distribution or whatever of opinion and you'll then will it be informative um in and so Norms I assert for these purposes that their behaviors with uh behavioral expectations that are set through the past what was normal and one of the assertions I make is principles and again they're assertions there's different overlap here the principles are institutional assertions of aspiration and ethics also are aspirational but they're human assertions about human behaviors that's the distinction I make whatever words you want to use and so it's interesting in the context of what you're talking about about querying a group because there's different evidentiary B group um uh chunks that you'll look at right normative you would be looking if if according to those definitions on hey what' we do before and what we do against the set of standards that set the principles may be more aspirational and it may be you know how we doing towards that thing but it's not necessarily based on what was done before you can you know you can say our principles are more aspirational than what we did before so it's interesting just that interface of what's used in the discussion of the group dynamics and how that relates to what you were just describing which is the querying of the group and individuals the framing of the nature of that output you're getting from the group is kind of interesting based on that question asked right what do you do before is one question what do you wish to do in the future is a different question yes so that query could be what do you regularly do so again it could be the person in the office and someone says what is your morning routine that's something we talked about in a previous week or it could be a query to a team what is your onboarding routine or how do you welcome new members um and then this question of the aspirational and the Norms so it's funny because Norms can mean what's normal it means what's expected that's what has happened in the past because what's not normal you know if not if not the past what's normal but at the same time normative means uh what should be so a normative claim should be like I think people should be treated this way and that may or may not reflect what has been the norm so people all of uh belief systems make normative claims that sometimes are consistent with norms and sometimes are uh almost radically Divergent from Norms someone says the norm is a but I think the norm should be B so I'm making a normative Claim about the norms and it should be different and so these are all interesting things to uh yeah Scott so one big gap that's out there in the world right now is the gap between law and what and interactions that technology allows and I'm talking about fractally up and down the chain reproductive interactions biomedical population wide all these different things the law is really lagging okay so one of the things that's interesting about this is there a possibility that this kind of analysis can help us to discover Norms that would guide um uh groups and those Norms if they're sufficiently interesting to the groups can be made enforceable into laws and so that to me this is a very interesting way of of a possibility of discovery of new laws and self-c constraints that groups can put on themselves in order to function as larger systems in the globe kind of context but I'll put that aside for now but I just wanted to raise that is is this a discovery is this a mechanism for discovery of new working models of larger scale norms yes Stephen yeah I think that that this is a good point because and and discovering that with this model here where you've got this task specific and then you got this effective actually as a as a larger sort of scale time scale sort of checking I think it's also relevant in terms of how we can get mechanized in our like if we just keep doing the same task and we get the reward and the queue and it starts to create a high sense of this false sense of certainty in our world which is so much about just constructing more of our social Niche and and alienating or externalizing the environmental so we we can start to get caught up in this kind of task specific ways of working and silos and so that maybe there's this this could speak to why we need to do some sort of sense breaking to sort of take us out of some of these mechanizations and look at a longer time scale um and use a effect to do that integration and not just rely on metrics cool let's let's take Scott's point about this being not just an enforcement understanding but a discovery process and connect that to some topics we've been considering like bottom up versus top down function of systems so the bottom up is learning that's where the little ants are going out and they're finding different things or each of the cells in the retina is receiving a different Photon each person is receiving a different Newsfeed and then as it goes quote up or in the system it's being aggregated and that is updating progressively more and more summary like variables like of teams or of organizations and then Norms can feed back down to lower levels of the system whether in the brain like the basing brain or the predictive processing hypothesis or whether it's team communication Norms uh and those can be informal or formal when we're talking about human organizations we gain this level of uh syntactic framing of top- down priors so instead of just saying this is how things are like uh the red blood cells are flowing at this speed so that's the speed they're flowing at we're going to say there's a speed limit and if you don't follow the speed limit then we're going to have this type of enforcement mechanism and so it could be the case Cas that in the legal context that if you just apply a law a top down law it's kind of like trying to teach someone a motor pattern uh let's just say a speech law of some type of controlling speech it's like controlling someone's motor Behavior again because it is a motor behavior and so what would you do would you put them in the exoskeleton and say okay here's a perfect golf swing now you're a golfer right well no you actually need to practice and you need to have feedback and so if the law is overly punitive and it says oh that one time that you got put in the exoskeleton you didn't understand how to golf and other sports that golfers know how could you not have made the connection there from that one time that I punished you for messing up that is not an extremely helpful framework for certain issues so how could we have this biral conversation where there's experimentation and implementation at the edges and then people's experience percolates up upwards in the system and then the system informally but eventually formally ends up entrenching healthy or productive or agreed upon patterns so it's an interesting thought about how these type of systems are uh not just about learning they're not just about exploitation as we've been thinking about they're sort of in both domains and they can trade off and when they need to exploit they can dip into exploitative uh type typ Behavior but also they can be in an exploratory mode Scott and it's also so much wrapped up in identity it's amazing how this is a new platform for identity because every thing has a Markov blanket and it has attributes and it external attributes that are presented and so the discovery when you have things like trade associations markets communities of Interest were things entities with similar um I guess marov blankets and attributes can cluster together forming those systems which have larger scale which enabled drisking and leverage for the organism or organization at at at larger levels so the discovery process of bringing these markets are where Markov blankets are um made similar right because you go in and there's a law it says if you're going it's a flea market your table can only be 5 by five so if you go in with a 7x5 table you can't go to that flea market you have to follow the rules to be part of the club but if you're in the club you get the power of having more people walking past your table to sell stuff so there's the interface the law for the vendor to the flea market and then within that transaction there's a subtransaction which is if you want this little pocket knife you're going to give me $3 and then I'll give you the pocket knife so these are the nesting of agreements in the context of identity and forced and unenforced Norms blue and Stephen so just to kind of tie into what both you and Scott were saying so like you know if we're ants and we're going through the world like building this generative model like as we discover as we're learning right like so as we learn we build the generative model like to what extent is our generative model tied into our identity like that's um It's Curious to think about and and I mean I know that like there's a policy update and perhaps the policy is more based on identity like we talked about earlier like the running every day uh maybe it's it's a policy thing that's identity and so that the we pick the the policy that that then minimizes the free energy but but I just wonder like you know the the generative model how frequently that that builds upon the identity like what the relationship is between those two okay and then Stephen yeah well one thing that ties into that question of identity and this this type of model is and the mark of blanket I suppose is how much changes based on the D the the the processing which could be um like this um in the brain could be sort of structured through the way the brain is structured so there could be an additional type of structuring going on in which all of this is able to work together and then so how much is is this kind of um the observations and the states in the sort of brain computating in relation with the body and how much is it just literally the the way the mark of blanket itself is changed you know so and I think that's maybe an open question at the moment because there's the question I don't think as much as the the idea of regimes of attention's been talked about but literally how would that change your computation by bringing in different parts of the body and the relation to the environment as the sort of Mark of blanket to integrate information coming in and how much is through computation and I sense that a that at certain times maybe when you're learning something for the first time it's very computationally heavy but then as you draw things down and it becomes kind of tacit knowledge we might become subconsciously able to flow and shift yes formal systems can become integrated through muscle memory so somebody who's learning how to play violin they don't need to know the math of the frequency overlap they might learn that in their Journey but also it becomes Incorporated through culturally acquired practices and muscle memory and that could be carried out into other areas Scott there was an article uh just read recently in nature or science which relates to the the um Steph's last point and it was about how in in cultures where there were Deaf culture so there isolated populations I think there was one in Central America there was one in the Middle East of deaf people because it was some kind of genetic thing and so large populations of isolated people who were deaf and what they found is when they came up with their independent sign languages they were all independently generated what they found in this research is they used the body in each of the isolated populations the body was used in the same way different parts like first I'm going to make up the parts because I can't remember them but they would first use the face parts then they would use the left arm then they would use the right arm then they would use the core of the body the morphology of speech through sign language was identical across the human populations even though they were isolated from each other in terms of what they added on I forget what it was with the bigger Concepts or the the things that were emotional there were different morphologies of that sit I guess situating their expression in their morphology in other parts of their morphology so I thought it would relate to that I'm not sure how but it feels like that form of expression being uniform across humans it's like the um theories of humans having a language capacity I forget who who's that is maybe mclan or somebody anyway but it's it's interesting that the morphology of the body sounds like there's some intrinsic um uh tearing of what Express ression comes through what parts of the body I'll try to find the arle very interesting and that actually reminds me of William Blake's poetry and the four-fold albon which is a metaphor for Cosmic Persona as well as all of England as well as the individual body and so it's a metaphor that he returns to many times in an allegorical context and it's so evocative because it speaks to a multiscale system and it is of applying to multi multiscale systems Stephen yeah thanks for that uh Scott because I think that's I mean that could actually also you know there's all this thing about this innate language capacity which has now been kind of discounted but I mean if actions first and the morphology is maybe more related certainly within certain you know environmental niches that people have grown up in um that would maybe explain why there was that confusion in the language world because people s of Saw language Universal well actually if the morphologies where it's at and language is just an another type of morphology that's kind of building on the core morphology of the body that's where that pattering was coming from not the language itself or the language in process look and I was just going to say extending that out further one of the assertions that I've started to make and and believe is that the mind doesn't reside in the brain at all the Mind resides in language and in material culture and the brain is an antenna that we tune into the local mind wherever where we're exposed to that's why colonialists put indigenous populations children in their schools right they take they want to tune their mind into a colonialist culture early because they then renders the externality innocuous for their colonialism anyway yes that's true of all children's education uh including cont temporary and to bring in this key point of precision um again active inference is not about systems finding the most rewarding State and then sticking with it it's about Precision so to connect that to Identity um the child's education is being uh it's basically being given symbols that will help it navigate the regularities in its Niche and again there can be dysfunctional education that leads to children that are incapable of taking advantage or of working with the regularities in the niche but this is where the policy selection piece comes into play to kind of return it to the figure because that's like important to return to what the paper said is the policy selection piece is so tied up with identity it's not that there's an identity like who we are and then there's this totally separate question of what we do it actually is we are how we act and to even take it to another level the understanding of how the world changes s the states that are external that we care about the understanding of how those States change is related to the policy actions that the system takes and so that even brings this relational mode another level deeper it's not just like saier War oh well the words that you use change how you think or they influence how you think it's everything is influential on how we think because how we think is like this emergent outcome from all those factors so it's a starting point to say that the words that we have or the affordances or the sensory capacities that we have shape our understanding of the niche that's the starting point blue so just to piggyback on the idea of a team and identity like like how and also where the mind is like where is the mind of a team right like when you're when you're thinking about this in terms of active inference and so the the individual team members I mean we've all worked on teams before at at various points and it's amazing how the team changes based on the like the function of its individual components right like so there there are different people that bring different Dynamics to a team some people may be louder stronger or less influential or or everyone has their different kind of role in a team and and these all like really bring changes to the overall structure and the way that the team will react right like like based on a query like when you query a team you know what is the best project of the Year that'll change based on the the individual components of that team yep and the team organism mapping let's think about team onboarding and the way that the interface of the organism is like an interface of a team so the immune system of a person there's times where something is onboarded but in a very controlled way and it does not trigger the immune system so for example if you eat something in the stomach which is a specialized chamber that digests food it is then absorbed through the interface of the gut in a controllable way in a way where only the safe pieces like you know that sugars are coming in and the amino acids are coming in lipids are coming in so through that controlled and evolutionarily selected upon interface there is an onboarding process that's healthy but if there's a onboarding event on your body like a cut in your skin it's going to trigger the immune system so that's more like a security breach for a team whereas eating for a team might be like hey we're going to get this ingest of new data our colleague is going to send us a data set it's going to come from this email address and it's this type if it's from a different email address I'm not going to see it or if it's of a different type we're going to need to change our pipeline but if it's this type of food coming at this interface and it's you know this structure we're going to be able to digest that and then use the information latent in that input to help our team become more structured or to expand just like an organism would break down the food and uh use it to grow itself or to operate Stephen and also with team you've got the the teleology of the team like the purpose the goals and I think that that I mean this figure sort of speaking a lot to task based you know ways of of things being evaluated with rewards and the way that we structure our teams often with the way that we then construct the environment to sort of lead us towards those rewards in a way and I'm really interested I've been looking at this thing called infusion space that I've been working working on is when we are in a kind of an organized space with a teleology and and a direction like a team and organization and then when we're in a sort of community uh space where which maybe this isn't I think it's another process of affect where we're sort of attuning to um that kind of inter subjective dynamic between us and you know there's there's one's more efficient the team's more efficient if your goal is the right goal the danger is that you could be they talk about now with the modern world is we're getting very good at doing the wrong things so it can be the Trap you fall into so climate change being a classic case we're very good doing things we shouldn't be doing um so I think there's an interesting question there about a teleological Dynamic aligned to goals and rewards which everything's then and could actually remove the The Knowing of your affect and you can just ignore it because you're getting other kicks and when the deeper knowing is like you know it's not really going to be good when you talk about really long time scales okay let's expand on that so thology is coming from the word Tios which is the end so tileology is the study in philosophy of what systems are as defined by their ends so the purpose of something like houses are to keep people warm and what are people for but instead of engaging in the philosophy of heliology like what is it mean to be a human what are we here for we can think about it from a cybernetics perspective which is also equivalent to like an intentional stance in philosophy and that's kind of like saying like Okay look whether or not we're ever going to figure out the ultimate Tios of a human or the ultimate Tios of our team at the very least as a goal-seeking system that has to operate amidst uncertainty and communicate through limited and noisy channels we need to have Clarity on our mission our local Tios so that we can have Clarity on action and again if your local mission is something that is uh pursuable then it's it's a second layer whether it's quote good or bad and that's the whole question about what Scott raised with the marov blankets of uh legal entities and there it's kind of easy to see some of the analogies between let's say a society and a person if each cell were to fight for its marov blanket then things could get hairy pretty pretty fast whereas over evolutionary time the interfaces between cells of the organism have evolved through so many different proteins and so many different mechanisms to work well together to be compatible within the organism and similarly within a healthy operating system for legal entities you could imagine there would be healthy interfaces but in the absence of a healthy interface then even entities whose interests are completely aligned might end up in extremely adversarial context Together Scott and I think that might be the problem with kant's Emanuel kant's uh what is it called the something imperative categorical imperative yes and is that the notion that if the a human acts a good in human acts in a good way it'll be good for society so uh the the contient imperative as it's uh called a categorical imperative sometimes uh it's uh I'm just just wanted to look up an exact word it is quoted as a rule of conduct that is unconditional or absolute for all agents the validity of claim of which does not depend on any desire or end that yeah that reads heavily as that classic era of abstraction in philosophy and that's one of the problems in European law right now and identity law like gdpr and those kind of things in the real world their idea of entities and of identity is based on Hegel and kantt and those are there there is a real focus on that individual being good and that being enough to have the entire Society work well and I think that's part of the pro challenges over there that that in Continental law you make the law um and you refer back to the statute all the time in common law which is any place that was part of the British Empire you what you do is you come up with a half-ass version of the law and then you allow common law to fix it up so the common law acts as a continually Dynamic revisiting um because the courts then make the law and then when you when you have a later case you don't just look at the original statute you look at the cases as well the other court cases and so it's been modified by that so one of the things that this gets Tangled Up In is philosophically the question of how much of a teleology you have I believe my personal belief is that because they had royalty a tradition of royalty on the European continent that they're more amenable to a single voice and so that single voice can act in a misdirected way for a longer period of time because it projects onto a paradigm onto the population but it's just something that it's it gets softer at that level perhaps because we don't have the data and the metrics but it feels like those same kind of things are iterating from a system perspective if you have something that's too stuck in the mud it's not going to change as fast and now that's happening everywhere since technology mors law increases interactions exponentially all of our institutions all of them were intended to drisk and leverage much more leisurely institutional context so they're no longer fit for function in the interaction Landscapes we have and so one of the questions is does how in what way might this analysis help us to not just improve old institutions that might not be possible but construct institutions that can better be dynamic with the increased frequency and volume of interactions we have which is perceived by any one institution as an increase in density interaction density that's why your calendar is all jammed up because we're still using calendars the way we used to use them which is linear progression of things they have to do things like that so anyway sorry about the question there yeah interesting Stephen well that might that also might tie into this idea of when we have systems in our Niche that we've created when you know when do we need them to be more able to our own active inference types of Dynamics like if we if this is the type of way that we can you know what would help us being better at attuning and what types of attention what types of scale would we want this kind of dynamical process and when does it want to be fixed and it kind of speaks another problem in the development world was the rights based approach to development so someone somewhere defines what people's rights are and what's and then it gets dictated down but that causes a lot of problems when you roll out into cultures which value things slightly different to those that decided what's right and it's also what's right for someone in a poor context or a poverty context might not be quite is with someone in a in a in a developed wealthy context so the same you know maybe this is a question that needs to be asked is when do we need to have this kind of active influence or godic help you know thought about create and when do we say look this isn't being done that way we're just going to Bol it down and use a non-ergodic kind of structured state of this sort of laws okay interesting so let's try to connect that to uh cont and then back to action because that's the key piece and that's what we really want to emphasize with the active inference approach is that although we generalize we're generalizing to reduce our uncertainty about observations which could be letters or sounds we're hearing we're reducing our uncertainty about the stimuli that we're as people receiving by generalizing so that we can guide action and so the categorical imperative on Wikipedia it says it's best known from its first formulation Act only according to that Maxim whereby you can at the same time will that it should become a universal law so that's actually also related in a sense like a symmetry relationship with the Rian theories of Justice like what would be fair would be if you were to be a random draw you'd want to be in that Society versus another Society but here's where active inference and action and the reality of the finite world come into play different people have different risk preference trade-offs so maybe somebody says yeah I'll take a one in a million chance of being the dictator and somebody else says no I'd rather have an 100% chance of being in this other world where everyone has the exact same amount of official power for example but even at a deeper level that's a thought discussion and it's becoming increasingly lifted from action and from policy selections that we as the agents perceiving that question are being faced with and so when people bring in the things like it should become a universal law that's a normative Claim about a universalism about a law you know and who's the judge jury executioner so although the sentence the imperative started with act according to dot dot do dot dot it ended up in the clouds with some speculation about what could or would or should become Universal and ostensibly enforced by other people with weapons but not yourself so how do we tie it back to the real system and confront oursel with what we actually are being faced with from an affordance and from a challenges perspective and then ask what policies will help guide the whole system towards states that we want to see but at first we can just acquire Precision about who are we what are our relationships what are our interfaces how can we think in a more rigorous multiscale way about how we are composed of our relationships so not that I'm some Adam and I'm being influenced by my relationships but it's multiscale relationships all the way up and down so what is really happening here these are some interesting ways that sidestep morality when working together but they sidestep the ways that morality has been framed in terms of well I think it should become a universal law that I should get all the money and then it's easy for me to act within the categorical imperative under that understanding and someone says well no that's not the real understanding and then you're right back to square zero so how are we going to use systems thinking but also include aspects of Intercultural communication and understanding that people are going to be having different experiences of the same deal the same interface different people are going to have different experiences of it and then work within that environment so that's kind of you know some big areas that hopefully we'll be thinking about in the coming years because there are uh serious questions about how to interface formal systems with people Scott one of the things that's interesting right now in law is that you know it's really showing wear and tear and people are really focusing on liability and rights and things like that which is terrific but one of the things the right is an is an expectation and what the law does is it constructs a bunch of Duties around that right that give life to the right so if I say I have a right to free speech and nobody has a duty to respect it then I don't have anything except words on paper and so the duties the construction of Duties what that does is it's constraint on other uh expression other Marco of blankets right you're limiting and constraining the environment of other individuals you'll meet because they have a duty to do x a duty to there's a a judge who said years ago in in United States your right to swing your head arm in public ends where my nose begins right so you can keep doing this until my nose is next to where you're swinging your arm so it's interesting isn't it that you have this that is really that Dynamic element if a nose is introduced into that system you got to stop swinging your arm right and so the anticipation of situations where you shouldn't swing your arm don't swing your arm in an elevator you know that kind of thing right don't don't spit on the subway right you start to have these rules which are like hey we can generalize that duty to a population and if it's relatively innocuous like red means stop purple could mean stop it's not the light that stops the car it's the behavior elicited by the signal so there are a lot of innocuous things we can avoid avoidable harms by agreeing on innocuous signaling for certain duties and behaviors that would render those rights to be realized and to be live living yep so with this kind of eating team metaphor ingestion of stimuli or of food at the organism level and that's being processed either chemically with the food or informationally with the stimuli in order to guide action so we can think about these teams and organizations as getting inputs and then engaging in action in their Niche and the goal of the inputs and selection is going to shape this is so that the inputs can be processed in a way that leads to outputs that are effective and so again another mapping would be to a heat system so like an engine if you can convert 100% of the potential energy into work it's a very efficient system and it doesn't lose a lot to heat whereas if you have a system that's broken or has inefficiencies in it it will perform very little work in its task specific uh role and it will release a lot as heat which is like wasted energy that's like data exhaust there's wasted data there's so many mappings here actually in the last couple minutes let's just run through the slides see if anything comes to mind but other than that I'm sure we could keep on you know thinking about this figure two of the paper the authors showed this roll out game and uh the agent at each time point is imagining what could happen on future timelines and so here's a timeline where I stay in this neutral State on the left side here's a timeline where I get to state three and that's good and then I go to state four and it's bad and so as you simulate and you roll out more and more and more deeply that's the sophisticated or the Deep component as you roll that out in order to track what uh kind of a world you're in are you in a world where all the routes are bad or all them are good or some are good and some are bad the agent is tracking a few things it's tracking the state estimates like the expected value of different policy choices but also the variance or the uncertainty and what we saw in this paper was that as the agent starts initially thinking and simulating deeply it starts to be perceiving it self in states that are rewarding and it has relatively High Precision about being able to maintain itself in those States but as it continues to simulate deeper and deeper the Precision of its model drops this is associated with anxiety from the computational Psychiatry perspective and uh it's also associated with the increased percentage of the events being negative but not a 10x increase in the events being negative and as we pointed out last time it's actually when the threat perception is at a medium level when the anxiety skyrockets and again this specific Trace is just related to the parameters of this model so it's not meant to be a general Claim about anxiety or something like that it is for future work to really map this onto Real World Systems we had the formalisms which people can pause and look at um the formalisms of sophisticated inference and the uh real natural language translations of some of these variables that are interacting in these papers we talked about the generalized free energy and about how that formalism has a few different parts one of those parts is conditioned on policies and it talks about the relational mapping between observations and states and then the other function here is conditioned on policy and it's uh yeah sorry let me just rephrase that one of these sides is conditioned on policy and relates to States the other side is conditioned on observations and states and it relates to policy selection um we talked about a few other topics intrinsic and external value General and special cases and uh how when you are moving through a prediction in this branching way there's an exponentially increasing number of states and so in order to deal with that organisms or systems have to engage in a little bit more of a guided kind of search and that brings us to this slide which is what I kind of wanted to pause on because I think blue this is what made you say we have to invite Scott so what what was interesting to you here what made you think of that just uh Scott and how um Scott's always presenting on risk and this just was like well I think that Scott really needs to be here and based on also the fact that Scott um said he'd been studying active inference since the October complexity weekend so those two things kind of nailed it for me it's I feel like it's a it's this giant thing smorgus board that's in the Next Room that I haven't yet entered into that's it's it's so delightful I'm so thrilled by my ignorance in this this is the best gift I could ever get so thank you all you know it's like a roomto room party you just go room to room and it's always a new setting but we're bringing the party along because it's enacted it's not that there's a room of researchers or a community that's waiting for us to be on boarded onto we're we're co-domesticating each other and creating the community that we want to see so that's self-organization and multiscale organization in action and we can think about that actually tying it back to this figure so the active inference framework under special cases which were described in this slide we can think about a few different cases where certain things are driven to zero so when there's no uncertainty about States in other words states of the world can be perfectly known then risk sensitive policies can be enacted so um when there's no ambiguity about States you can talk about risk that's this part right here however when there is ambiguity about states you have to reduce that uncertainty and that's sort of like if you don't know what the states of the system are you you can't operate but to the extent you have reduced ambiguity about the observations in States you're able to increasingly plan risk sensitive policy so Scott that makes me think about in a world of digital transactions where we can have reduced uncertainty about certain types of events happening could we then move some of that energy and focus to risk aware policy so that's one piece here that's why I call what we have right now is self boiling nice so that's the that's the saddle meme this is still within the context of how most people think about ambiguity and uncertainty vuka that's in the terms of minimization of two different things minimize uncertainty minimize risk first go ahead and reduce your uncertainty about the states and then to whatever extent you can do that reduce our uncertainty about the future Vis A reducing risk that's the normal framework for most people or at least it's one way to think about it but now let's think about it um active inference not just in a risk control way but in this basy and statistical framework a variational framework where we're actually doing not just a double minimization but we're doing like a push pull with a minimization and an exploration simultaneously so we're going to be able to dynamically adjust explore exploit it's not exactly explore exploit as we talked about but it's like explore exploit because we can dynamically adjust the relative pressure waiting of the in and the out so we're always exploring and we're also always reducing our uncertainty about the system and so that's why the uh two arrows are moved into an overlapping uh area just graphically this is just a loose graphical model here because not it's not a disjoint two-step process ambiguity about States and then ambiguity about risk if we have certainty about States this is the co- joint estimation of all these pieces together in a way that's a push pull so there's a couple Paradigm changing ways of framing risk and operating in this uh mode and that will it'll lead to different types of institutional structures that'll be distributed which will resemble neighborhood watch but it won't just be neighborhood watch it'll be neighborhood creation as well that's a fundamental change exactly instead of just putting up the law at like a trip wire and then triggering it you could say how do we just have people who are going on a neighborhood walk just people walking the neighborhood it's like you know like they could just having fun and being social could that be the actual thing and so it fuses several of the previous ways of thinking about it well we're going to reduce our uncertainty about what's happening in the neighborhood and then we're going to do a risk minimization policy well this makes you think about installing surveillance cameras but what if you said we're going to jointly be discovering what community Norms we have and what we want to be aspirationally as a neighborhood Hood around safety or around communication and maybe there's a way where we can just go on walks in the afternoon and all of a sudden people are connecting and it's a safer world and the other just one other thing I I once read an article about the world champion fly tie tire and he tied flies and they were the be most successful flies for catching fish but all the people complain that his flies didn't look like actual insects they didn't fool humans and he said I don't have to fool the humans I just have to fool the fish and so the reason I say that here is the way we're going to do this is we're going to use a left side we're going to get people in by saying hey you got risk what you been doing about the risk let's talk about the risk okay that's fine we're going to get other people in the room who are going to be talking about the same risk but they people never met each other before and then by the time they leave the room they're going to be in the right hand side of the design that's where the governance is going to shift we're going to hook them with something they res they recognize like The Fish Guy we're going to hook him with the thing on the left which is hey r we're going to talk about risk that's with the atlas of risk I'll send it around to the other folks but some of you have it it's 870 different information risks that are shared by people now because digital stuff has made everybody have trouble so they they come for the the risks they know but by the time they leave they're going to be organized into different groups that aren't going to be dealing with the risk the risk was just the existing risk description was just a way to get them in the room yes yes so one metaphor before Stephen instead of let's let's measure your blood measure your blood measure your blood so that we can measure your risk of the disease this is like how could we be doing active inference on health and working towards Health in a way that yeah maybe we'll do a measurement here or there and we do want to reduce risk and we will reduce risk but we're actually framing it in a way that's like a journey related to health in mind rather than ambiguity risk disease negative negative negative uncertainty anxiety how about yin and yang with a little bit of a more balanced understanding about how we can function and enact policy Stephen this is a very important diagram actually like you're saying this this takes us away from just trying to reduce uncertainty about the external stimuli and the kind of locked in risk aversion and you're actually working more with that time dependent or time series um approach that active influence gives of like well how are things changing and how just like that fish looking at the the fly how does that um embodied agent and the wisdom and recognizing that there's a wisdom about how to be in the world and how to be with your Niche how is that actually being used to inform how we make things happen and not just this more mechanical approach yes and good points there um in active inference stream 19 Yonder in mid April uh we're going to have deeply felt affect which is one of the papers that was cited in this paper but we're going to go into it in more detail we'll talk about perception about anticipation action metacognition implicit metac cognition and just to close out I listed a few questions um do people want to hang out for a few more minutes and talk for uh a little bit yeah sounds good but I'll just give the closing slide and then we'll return to the question slide thanks everyone for participating the live viewers uh have a link to a feedback form in the event everyone else really thanks so much for 2020 super special and unique experience for all of us so really deeply appreciated for those watching live or in replay or whenever wherever it was great times so thanks a lot and we're just getting started so for a little bit longer whoever wants to stay and if you want to leave just just no worries just drop off but um what's something that somebody wants to go into here or what's something that still people want to bring up one of the one of the things I'm wondering about is just the we've been taking this from a single individual perspective to some extent and in those group perspec Ive is kind of it's interesting that the simultaneity of all of it that the you you know you simultaneously have an individ it's this the how do how do we get a handle how do we help people get a handle on the the many levels that operate simultaneously and then one of the then then how do we bring that to the current time the now where everyone is acting in this moment I you know reading another paper now on fractal time and they say hey the past and the future don't exist if they do where are they show me right so the idea is now is all that exists so we have these priors which were that and we have these future projections which are that but we have the now and we all act in the now and all these Myriad Dynamics feed into the now and so it's such an interesting notion to me of how to trim tab systems as a lawyer you don't just we don't just observe we're always trying to push things in One Direction or another for clients right or for whatever and so it feels like it's such an interesting thing that we have it's not just a now we have a series of Nows available to us I guess because we in the diagram you see a series of Nows right in the in the steps so it's interesting to that temporal aspect is something that's just not brought into so many systems right they're so static in so many ways and so in a sense like when you working with standards people they'll say oh it has to be robust so that we can plan based on it but it also has ref flexible and it feels like this naturally this analysis naturally lends itself once it becomes an institutionalized thing because our institutions now are not built on these analyses but once they're built on the Dynamics they're built on I'll use the word uncertainty but it's really embracing uncertainty or as I said in 2014 you called it entropy engines we were actually mining disorder in order to create order so anyway just a couple of things that thoughts that came to mind it feels like that time issue is relevant to our understanding how we can mind disorder cool Stephen yeah the time issues definitely a big part and we see that with these papers we' seen that there's the narrative approaches there the reflective and now they're trying to get into the pre-reflective more and traditionally that's been you know meditation but what is the way that we understand these different time scales and I what I'd be interested in is is is how does this link in with affordances I think there's something really interesting about that Niche and the affordances in a niche because it it gives away because it's sort of permanent before you know like that the path could still be there the cups can still be there you yet my interaction with it can vary so I think that that's a a really interesting um way that experiments can tie a lot of this information or knowledge together sure so one idea we've talked a lot about is the idea of attracting sets or attracting States so for the perspective of your body with regards to temperature you want to be attracting back to core body temperature excursions outside of that range are not just bad they lethal so attracting states are really important and just like you can have attracting sets of States on a single number line like temperature you can have in the wandering itinerant dynamics of thought you can have attracting thoughts and you can ruminate and you can get too attracted or it can be too precise but it's an idea that people could be familiar with or not that attracting States can be thoughts so let's think about active inference the framework as an attracting State as something where we can in a room bring up these ideas and refocus the discussion or potentially expand or uh zoom in so we've brought up this multiscale and the Deep past and the Deep future so there multiscale through space like the nested systems and then there's multiscale through time there's molecules vibrating and then there's longer term things so let's start with the multiscale space and time just being our starting position and then active inference goes further it says not just is it going to start being multiscale in space and time but there's going to be nonlinearity and uncertainty because there are causes in the world so RJ and his alarm clock and his dog and something that you wouldn't expect changed in the world and that changed another part of the world that you basically wouldn't have expected um and also our action is always going to be constrained by our situational affordances in the niche so to Steven's Point there which is like it's all good to think about objectively what's happening but constrained by affordances in the niche that we're actually in is the operating conditions we have so it makes sense to at least ground the discussion in the affordances and then we're presented with attracting set of questions like how will we work together to be agents in a system of the type we would like to see that's a question that in various settings people could say oh well we don't have Clarity on what kind of system we want to be in we need to reduce our uncertainty about what kind of system we're in that's visioning and clarifying Mission but then there's other times where people say yeah how do we be the Agents of this kind of a system that we want to be in so we're not uncertain about the mission about the end cybernetically but I'm just like honestly confused about what next role I should take in this system to make this happen according to this mission that we've laid out and if my actions are consistent with the mission as we've laid it out then they're good but if they're not then they're just not helpful so we can start to combine a lot of these ideas and realize hopefully that whether we go deep into the physics side or into another aspect of active inference we'll use it as an attracting set and as this range in Phase space where we can bring the discussion and help ground it in the multiscale insights in the nonlinearity of the world and a few other things but I think it's yeah and they really are they really are that like we we start our diagram with entities that have priors right we're not starting with I mean you could start with no priors and you're just wandering around in the dark in the woods but we're starting within system that has something that we comparing the reality to right so they're trying to minimize the difference between their expectation in reality is that the free energy notion right so um so sculpting expectations at the front end is what education is an experience right and so with you know when we can you know our our system is our educational system in the United States is set up for after the based on the Prussian school system right that's the um D the deschooling society book by um illich um so we're our expectation is industrial workers well now we don't have industrial workers only we have service workers and so our educational system is not serving our folks for instance right so we have these um th those influences it's the I guess the the thing that's the thing I keep coming back to it's like a situated cognition that we situate our cognition in externalities well those externalities also have are entities that have Markoff blankets you know right now we're thinking together right this is like Plato you said you know dialogue lets us query Consciousness so we're doing that um and each of us has our own view on on the direction of this thing right but they're so we're not interacting with static things we're interacting with Dynamic things that have their own Dynamics and the recruitment of those populations is I think a lot of what happens in legal Concepts it's what's it's what right now I think this this the things we're doing right now could make a big impact on ethics discussions because ethics anything that's not written down in law but is bad is now called an ethical problem right it's just ethics is carrying a lot of water right now for anxieties of people generally with the Technologies and exponential increase in everything so it feels like we have an opportunity for intervention because there is a vacuum um of meaning in the current institutions and it feels like this kind of explanation might be a nice way to get a Toe Hold there let me build on that Scott so there was a book um I read called Legal systems very different from our own by a fellow with the last name of fredman but it is not myself or relative directly and it really does a great job of capturing how different legal systems from uh informal like criminal or prison legal systems Justice systems using the term quite broadly to different world traditions very very different processes that they followed but also they had different ends in mind now the current operating environment not just are we being confronted with different uh interfaces mixing that hadn't combined before but also there's new challenges and so that's sort of like there was a you know a two-stroke engine and a two-stroke engine and a two-stroke engine and they all were kind of coupled in a way where they all were working internally and now they're being cobbled together and there's a lot of heat informationally so to speak there's a lot of parts that can burn because there's so much heat being wasted with the interfaces and so can three two-stroke engines be simply rewired into a six-stroke engine well no it doesn't really work that way how can you link those engines together into a system and that's really the question we're we're um addressing here is narratively linguistically mimetically but just formally as well how will these different systems get C together so that we're actually connecting the gasoline to the car moving in forward One Direction and in one piece not taking gasoline and making something that's just combustible but only hurting people Yep this is this is where I suggested people read flatland because it's the adding of Dimensions right you add Mr sphere comes in and lifts Mr square up and he can see through you can untie a knot in the upper Dimension without crossing the knot the Rope right so in law what we do is you create a new dimension so there was a time when we didn't have credit cards and people used to do run IUS at the bar and things like that it just didn't exist and so credit cards now serve this huge function they drisk huge sads of interactions right but at one point they just didn't exist and this guy de Hawk wrote a book called the birth of the chaotic age and it's kind of a crappy book but it's kind of an interesting notion the banks were like no no no no no we don't want your credit cards go away get out of here was a paradigm shift and then eventually they got the credit cards got in and now try taking the credit card fees away from the banks right so what was impossible becomes inevitable and it feels like because things are so stuck this kind of approach that we're talking about when taken in markets and institutions is going to that extra dimension of credit cards or swaps frees up it allows for these interactions to take place just like the extra dimension in flatland allowed for Mr Square to get New Perspectives on his life and his existence so it's it's a really it's a fundamentally different and that's why they it's so interesting the Markov blanket nesting gets really interesting what which Marco blanket can Nest within another one and when you have community of Interest one of the things I always remind people of in in New York for instance you have Little Italy you have a Chinatown you have different areas where people go together because their markof blankets their language is easier it's easier to communicate lowers the cost of communication lowers the risk you also have the jewelry district the the Flower District and the U cookware District economically people cluster together so communities of Interest they're collocating well here if we look at your diagram with the four states you know the two states on the left two states on right every one is now accumul they're going to communities of Interest based on the Dynamics or rigidity of that left-and two states you know oh do you share a risk do you share a solution but what if we could have communities of Interest accumulated based on their capacity for change instead or something like that just to raise one really important piece then Stephen is when people align around on the left side of that figure reducing risk or shared policy or shared culture it's often a similarity clustering okay so people often cluster based upon similar needs similar understanding of the world similar affordances but let's imagine um the reality of the organism which is that actually it's a clustering on the opposite in some senses of disjoint abilities so in this context of the body if you said well there's one kind of cell it can't move it just holds a bunch of fat and it it literally just does hormone and fat processing and there's another kind of cell that will just dry out in second in the Sun but it can contract once and there's this other kind of and it's like you'd say they're not going to live in the same neighborhood together it's like oh wait no they do that's the body and so how are we going to interpret this all the levels of what it means to be a person economically and culturally and psychologically and understand that it's neither simple Affinity nor simple disaffinity on any scale that will lead to function because function and the islands of function are rare in the almost unimaginably large State space of possible legal codexes or possible programmingsoftware system it's not a simple imperative but at the largest level that accomplishes our mission so just such interesting topics Stephen yeah and this well just like you were just saying there about these different cultures and the niche construction I think this can help there's a lot of problems in terms of discourse around this not more broadly in the world and I've like when we talk about systems out there in like City but then well how much is a system how much has been an evolving Niche construction some of which is constructed in this kind of linear you know this kind of teleological strategic way from above and and and geographical way and um but also some of it's just because that's the way it's evolved over time with some of this like we say these Dynamics I think haven't really been brought in or if they are brought in they kind of ends up in the s and it it all gets framed in whichever metaphor a particular sort of school is applying and that you can get trapped by that and and I think this this actually came up for me today so just like you talking about the indigenous um ways you could have of governance or other ways is I think that we we have this idea of systems and these different systems and how do we integrate them and this this assumption that they we need to be integrating them but like the these indigenous systems you know it flips from one to the other so for instance we have our Western system which seems to be like and we need to attack our system is quite a sort of a progressive leftist approach or we need to all revert to some sort of native indigenous system and they had it right and the body knows everything and if we just listen to our body and there's this kind of Gap whereby how do you find a way which is kind of maybe configuring different approaches so it's like okay well where can we configure the other ways of knowing other ways of governing other ways of and it's not just a case of like mash them together because they may just have a completely different dynamical process and then they don't integrate they just do different things at different times like cells do in the body of course it's all going to come down to the details so you know none of these metaphors or um you know questions are even the beginnings of the answer but yes that's definitely where we want to be understanding at that the level of societies is how much formal integration will there be now if somebody says I have I designed the interface for different operations that is still coming from a cultural context so the challenge and part of the quest will be to frame that interface in a way that doesn't pretend that it has the neutral Arbiter ability cuz that's actually one of those Universalist fallacies that were fallen into by abstractionists now we want to be grounded in action and still talk about an interface for action and so I think actually that's why the task may be easier than the thought bringing together could be the action bringing together I'm always reminded of Scott's example that he told me a long time ago with two lumber yards that are competing in the market for selling Lumber but then they have the same insurance and then I like wait they also could have the same internet provider and they could pay taxes to the same local government and they could have the same lunch service it's like everything except for lumber so it's like how many fronts they might even share a fence they so many areas so they're almost collaborative on more fronts than not and so how do we take this understanding that on overwhelming number of our edges and even for each Edge on most of the domains there's just massive areas where we're in alignment and then risk can be catastrophic so it's like if we just reduce the amount of cascading failures in the system by playing to our strengths in the areas where we are aligned maybe all of a sudden now the lumber yard people it's like oh you like this kind of wood I like this kind of wood we should just Niche partition but we're not going to go Niche partition and fork and never talk if you're going to be selling the luxury and I'm going to be selling the consumer wood we need to have a better communication interface oh you have employees who speak this language and I have employees who speak this language well let's make an agreement that we're going to send numbers back and forth in this file type on these dates and here's who we're going to go to if that doesn't work it's like wow these two companies were fighting on the market in a previous or a different Paradigm they might have you know been wasting money on advertising or committing different sort of Gray Zone interferences with each other's operation but just by reframing it in this multiscale system even if you're not doing a government change whatever government system there is those two lumber yards now have a new way of talking to each other to their employees with their lawyers whatever potentially in a way that could lead towards them working together better so that's just very powerful I think that's what it will be cool to continue returning to and and we're moving there really from a a zero sum game to a non-zero sum game right there's a winner and loser and what you do that's why you add I always think of that you add the dimensionality and it recasts the game as not a winning and losing game and that's what we have to do in GA Theory right if you want to save the world from from us we have to recast the game as the is as the entire game right the the sustainable development goals even the UN has them they're even tripping over them because they have some they had one power plant they developed the garat power plant in Bangladesh and and they scored big score on energy sdg and on development SG and then they got sued by the local fishermen in the local population for pollution which was screwed them on the other SG they didn't even talk among the SGS at the UN now that's not surprising if you know the UN and the institution but it shows you that it's even it's hard to integrate even when you have the the same source of the Norms or the or the aspirations they just go for it on one of them and they don't they're not talking to each other right and here's why preventing failure is so critical um which body is healthier each organ system at 90% function or most of them at 99 and one is at zero well y that body is dead so all of the functions need to be adequate or better and we need to improve adequate improve resiliency improve accessibility Etc but preventing failure uh cannot be uh just part of the discussion on reducing risk reducing cascading failure has to be an imperative and so in that framework where we want multiple components to be functioning it just reframes the oh should the government be interfering here or should this person be undertaking this policy and again like I said like the task to synchronize on action might be easier than thought so those two lumber yards getting them to go to the same religious institution or send their children the same education or listen to the same YouTube channels why why even go down that road in the metaphor when you could just offer them a service where they're drisking their Lumber business so instead of saying we'll synchronize these lumber yard people on this and then they'll just act nicely together it's like yeah or you could just provide the interface so that whoever they are however things change they'll be able to play nicely together instead of just saying well if we synchronize them in this way then they'll know what to do they won't because people don't well and that's why one of the things that I talk to people about is I say our recruitment that I'm doing in the program at the lab is not based on on selflessness it's based on self-interest and the assertion is that they can drisk in ways they cannot do unilaterally period that's what I tell people and so you get together so it's a commitment if they want to drisk in these new ways they can't do it by themselves just like a stoplight I can't I didn't send out around a memo saying let's all stop at the red light that was done right these are these are scales that we can't do unilaterally and so that tempting that bait of saying you can go over there and like with swaps Australia swaps were outlawed when they first came out and the banks in Australia were getting killed because they couldn't swap out their interest rate risk so they lobbied the Australian government to allow for swaps and then they did allow for swaps and they were able to swap out their interest rate risk like other Banks the downside is they all then got hurt in 2008 when the swaps were used for speculative purposes that's a different problem but that invitation to drisk in ways you can't do alone it's kind of it's It's a crass way of getting people in the space but to me it's the hook that joins that left hand side and the right hand side with those four spheres because you right on the left hand side you have this characterization of risk as a zero sum game on the right hand side well maybe it's not a zero sum non-zero sum in that context but it allows for a bigger the dimensionality of drisking increases when you increase the system size so if you get those people who are similarly situated together it's new markets that's what trade associations are right trade associations people are competing all them potato chips Lumber it doesn't matter trade Association competitors get together and what do they do share information and insurance those are the two biggie things they co- insure for some whatever risk they share andless they can buy insurance cheaper of whatever kind and they uh and they also share information so they can understand situational awareness better yep I put up the slide here again um yeah Ivon so you want do you have anything you want to add or or remark on no just chilling yeah okay continue Scott wait weirdly I think we just lost Scott's audio in between oh no I okay okay sorry continue uh but I was thinking I there was some other thing I was going to say but I can't remember now about that that oh yes yes this is an example from the Fisheries industry so M this Elanor Ostrom um worked very closely with people in the Fisheries so a lot of Fisheries are designed with Ostrum Notions and um so uh the of the commons and one of the things that a friend of mine who works in the um uh hold on a second okay Stephen any sorry one of the one of the guys sorry one of the guys who works in the um fishing industry he said they had a problem they have this thing called buy catch and it's the fish you're not supposed to catch so let's say I'm going to make up these numbers in the species but let's say you're allowed to catch 200,000 pounds of flounder but they an orange Rockfish or yellow rockfish and if you catch four pounds of that you have to stop fishing it's a protected species you can't fish anymore you can't pay your bills you can't buy the oil you can't send your kids to school you don't you don't have money so what they did they were trying to figure out how we going to protect orange Rockfish or yellow rockfish so what they did is they said okay let's pull the allocations of different Fisheries so if they had three guys each had a four pound allocation so they can pull it and make a 12- pound allocation so they can then what happens is if one of the one of the people goes out and fishes and catches five pounds of yellow rockfish he can still fish whereas he might in the past have had to stop so he spread the risk out but then what happens is the the other fishermen start to be policemen because they're cutting into their allocation so they go over and they say hey hey hey use a differ Siz net or don't fish in that area whatever they share information because he's cutting in it's an economic loss for all of them because they pulled their allocation so to me that's a big deal that that that idea because what they're doing is they're spreading risk and when you spread risk it's like insurance if more people have claims out there and my premium go up right so I'm going to encourage things that make things safer right as a as a stakeholder in that system and so similarly here it feels like when we move to that right side diagram we're going to have more Avenues with which to create information flows among the parties that will actually drisk better than the left side and make it cheaper Stephen and then I'll have a point on that yeah I think this this this use of this diagram could be important because yeah the in a way I suppose this diagram on the left it's like you either just put in laws or to to mitigate those risks and challenges or the transdisciplinary tool that we have at our disposal is money money is the thing that everyone uses basically as the transdisciplinary allocation of what to do on projects and it doesn't is it's Pro highly problematic and there's not really a lot of other things apart from what budget gets allocated that allows that so is there a way that this can provide us and and and at different scales because sometimes yeah it's fairly clean at kind of a national level or you know you have your rulers level and the individual but most stuffs happen in in between as we're talking about how can we somehow get a handle which is not too culturally metaphoric ly uh constructed but it's is is is maybe relatively Universal I know it's bad to use that term but I think there's something in that in in in being able to say okay this is how this information gains happening this is how this you know and now you're starting to have a way to talk about um

Dynamics um in ways which aren't normally possible yeah one thought on that and again to this point about you want to design the Nexus the interface in a way that is inclusive and accessible but respects that not just do people disagree but they have actually different visions and understandings about this whole system that we're in and so here's and then where we can tie it to this figure and how it could be Ed to communicate and maybe even organize around so on the left side it leads towards abstraction because it's like well let's let's wait until we're less um ambiguous and then once you have no ambiguity it's like well let's figure out out what the least risky policy is but guess what all the policies have risks and even if there was one that somehow had less risk than strictly all the others there's still Black Swan events and there's still people with different preferences for uh a given pipeline for oil or for a given fishing agreement so the idea that you're going to minimize risk for all parties in a way that they're satisfied with is going to be this Quest that you're going to be at the the drawing board quite probably forever let's contrast that with the right side and you know the the authors I hope will be okay with us taking wide latitude because I it's you know we're taking this figure beyond what y it's framed but that's what's fun right now let's go to that drawing board with the um the power plant and the Fishers and Ostrum and everything like that instead of just getting Frozen at the drawing board on the left so now we're on the right side and we're saying look we are at the drawing board so whether we're speaking through a translator or you believe that we have different interests we're here at the Nexus so Universal aside what's locally going to be working for our system well we have two imperatives we want to make maximum Information Gain but we want to maximize if you look at that ISO given P it's not Information Gain on the observations that's this part see how it's strictly in the O here's ambiguity about State observations okay this is maximizing Information Gain on States and observations given policy in other words words we're not just reducing our ambiguity about states of the world like give me a more accurate stock ticker this is like hey what's the most informative uh policy that we can enact to learn about the states so we want to query the markets in a way that isn't just getting increased Precision on viewing them and so now we're all at the table together so whether we believe ourselves to have or actually have adversarial interest in one dimension of 300 or all the dimensions or whatever we have an expansive maximization growth oriented goal together which is implementing policy that helps us understand our system and the mapping between the hidden States and the observations better together now by doing that we're also engaging in joint action which will prepare us for larger and better action and what are we doing joint action to do well part of it is to be consistent with the kind of system Who

We Believe ourselves to be but the other part the minimization what we're running away from the stick is disorder with respect to the observations so not with respect to us getting paralyzed about which policy but we want to reduce our surprise about our observations in the context of optimal policy okay so here how much CO_2 is there really in the world and how much do we really make and what would be the riskiest policy and should we just remove gas cars and I mean you could go down this Rabbit Hole forever right here is here's a predicted trajectory here's different types of models and what they predict about the trajectory of CO_2 levels through time and we want to reduce our uncertainty about that model together all kinds of people who believe different things we're going to reduce our uncertainty purely about observations as we jointly use policy together to learn more it's just such a positive way of framing it and again people are still going to disagree but we're coming to the table together so as long as we're not at arms and we're coming to the table together we don't need to go Universal we don't need to go culture free we just say we're here we're figuring this out today for ourselves now and for other people later and when you that's beautiful I mean I you're I can't wait till I have more facility with the concepts because what you're describing the processes you're describing are going to become governance it's not like we're going to find governance of the processes those processes will be governance that's going to how it's going to happen and so it's a very the question is you know we're we're on the left side how do we get to the right side what are the stages that we can do to move us over there how do we provide uh prepare ourselves as groups to be able to function on the right side in a natural organic kind of way one of the things that I've been advocating for is collecting practices from the left side but then having groups stare at the practices so you can say okay let's it's basically making different changing the assembly right the disintermediation of the left side has been done by the Internet it's done by covid it's done by the big challenges to those systems they reveal the limitations of those systems and so the disintermediation is done on the right side we're going to reintermediate because there is still mediation of the individual components in the system right they're brought together but we're reintermediate with a fundamentally it's it's fundamentally different in some ways in some ways it's quite similar and so one of the things that I'm interested in is trying to find ways to make it so similar that people don't even realize you know they or or they're comfortable with the move to this new framing of analysis very cool yeah Stephen and this this ties in as well actually this is so good that you've gone through this diagram it makes makes it very much jump out but this like the safe to fail approach of of trying to so if you imagine that policy wheel um the red wheel on the on the right hand side you know Act of inflence knows that that's really our choices are what we do with our policy we we a lot of what's out there in the world the hidden States you know we only have so much control over so if we're able to almost shift the policies in there and and knowing that they're safe to shift so these safe to fail experiments which is what they talk about with so if you do safe to fail policies which means that you can then observe the states observe the What observations observe the overlaps observe the Dynamics over time then you've got an awful lot of um new information that you can use and obviously the key is to do not to have something in there which will be um there this antifragile comes into that as well you know so you can learn as you do and and obviously knowing that there will be certain times when you got to pull away and stick to a you know risk minimization Focus but this you know it gives a way to talk about that probably to people from very different views of the world yep so another metaphor there with performance would be like a martial art and the safe to fail and the sparring and the play so there's times to play and you know for children and there's times to be serious but no matter what you're doing it's always serious because people can fall if you're walking you can fall if you're talking you can say something mean if you're you know holding data you can lose the data so everything is very important and so we need to have that resilience in mind one other way in which this can be used so this is um putting a little bit beside this uh purely like somebody at the table who doesn't want to communicate with us scenario so now we're in a situation where like people want to work together at least they've bought on to that level okay we're talking about the system and someone just has this figure up you know just as they do and Stephen you said it like what we control is policy we control climate policy you know what we don't control the climate but what do we control our climate policy we don't control biogeochemical cycles we influence them but we don't control the outcome and so people could be talking and someone's like well I want to see this okay you're talking about C you're talking about your preference vector and someone say or are we talking about the state of the world are you actually saying that this is how people feel or are you just stating your preferences about the world because it's fine if you want to you know I I want to learn about your preferences and remember your PR Es are the key factor that really go into our policy but our policy is not the same thing as the states about the world yeah so it's like okay now what are the observations someone says no no no actually it's 80% so okay your your inference is that this is 80% is that right well here's this observation oh well this study is biased okay well great let's you know what if it was a factor of two so there's so many ways just to have this figure up and someone says are you sure about that policy selection say Okay

W it sounds like up here it sounds like we're in the metacognitive World we're thinking implicitly about our own group's generative model of this policy decision making process and it seems like you're uncertain about this and what are you uncertain about okay what could we figure what could we reduce our uncertainty about is it your preferences is it the way that your preferences and our uncertainty are being combined with the affordances and the niche into policy that of course is the hardest not to untie that's control theory and everything but you can at least say hey is it your uncertainty is it your preferences is it the niche perception affordances not just the preferences but the niche itself e the prior and is that being combined in a way that makes sense to generate policy because if you if you disagree with the policy which part of these was it because that's all that went into it so don't just say you disagree with a policy tell me do you want something to be do you value something that wasn't valued by this policy okay now that we're talking about a given policy do we have synchronization at least cely on the starting State not on all the details but at least like the starting state for our scenario and are we talking about states of the world do you believe that our policy is going to influence the beat you think that the cutting the miles per gallon to 20 is going to change the way that the hidden State CO2 in the world changes through time you think it's going to do that well I don't think it's going to do that or I you know think it's going to do this instead and so now we have an interface okay policy is how states of the world are changing not just as a function of models of prediction but as a function of our interference in the system and it's going to be manifested in observables not just we believe that 80% of this will be happening by this future time but rather as a function of policy we're going to see these observables and if we're surprised by the observables we see along the way we're going to be course correcting and we're going to be at the same table again so there's so many actual structural ways to talk about this I think and and as well as narratively that people could even having this up maybe it could be simpler than this one maybe it could have a few more pieces but just separating out where people disagree it's like yeah in a legal I don't know yeah in a legal system I could imagine this being useful too oh well it would have to be recast as something that people wouldn't you know fall out of their chair because legal people don't like to see squiggly Latin and Greek symbols but but the but the absolutely no question I mean the way you described it what you're doing is you're really mapping I mean it's teaching and you're and and and and showing people you know whenever you have a map that says you are here you know when you're lost in the zoo or something like that and really it's a you are here map right so cge you know it's it's allowing more granularity and it's offering more dimensionality for Solutions right because if people are just talking about policy there'll be ambiguity if they're not talking about CG or E yes yeah what are your preferences C what what are the field of affordances e and the prior over those like the the quote nuclear option it's an affordance and it's the last resort so tied up in the affordance is How likely it is that's your prior over policies so you have the field of affordances and the prior about them then you have your preferences C I want to see a world where there isn't radioactive fallout and then you have your uncertainty and then those get combined in a way that is formalized with G and that relates to Pi policy selection being the outcome of the free energy being minimized which is again what's the most likely policy and someone's like wait most likely it's like right but most likely under the model of the world that you believe will happen not just the one that you think is like the run-of-the-mill because quote the run-of-the mill is not run-of the- Mill yep it's really interesting because I mean you're integrating subjectivity into an analysis of the reality in a formal way it has that I believe that this is where the Dual instrumentalism is the most powerful thing in the FP zone so the Dual instrumentalism again it's like there's it's also kind of like a Bailey and mot I don't know if that's being mispronounced but like where there's a strong and a weak form of an argument so again the strong claim would be this is what systems are and this is how they're um operating that is what a bacteria is that is what the Marco blanket is but the actual weaker claim is just we're going to use it as an instrument it's like we use linear aggressions to mod in Time series modeling to model GDP through time or regression of you know height versus risk for some disease it's like it didn't say people are linear regressions by doing that it did lead to a world in which certain types of conclusions were reached but it wasn't that people were linear models it was actually consequences so this is just a different statistical model and as we're exploring today and we continue to explore even qualitatively it helps structure messy discussions about adversarial relationships and in that way which we're only barely scratching the surface of like you know really formalizing the marov blankets around companies and things like that having this type of a structuring and being able to be uh disambiguous but it could be so much more it could be translated between languages it could be versioned it could have impartial um you know trust providing uh groups there's so much that could be done there and it's funny because so one of the books I have on the Shelf here is by hanaman it's called doing money and it's about how money is a universal risk consolidator so you know you get money and you can take care of your other risks basically so one of the things I've been wondering about is well money is kind of turned into Data because you don't really spend cash anymore so why can't data itself without being monetized be a universal risk um mitigator

Stephen you cracked me up when you said that money was transdisciplinary I've never heard you know people say that transdisciplinary research but I've never it's so true though the Departments will pay each other in money they won't site each other but they'll pay each other so it's so true and so and so that but data can data now be and can this system because money we we you go to an airport you got to convert one system into another so there's different encoding of the bills themselves right okay so that's but but ultimately I always tell people if I walk out of if I'm outside a courthouse and someone says I just got a $10,000 judgment I don't know if it's for a divorce case a slip and fall I mean it's money it's $10,000 is identical you can buy the same amount of groceries but it could be for various different things that you were you were compensated right so in data it feels like this model might allow us to have data be a medium of exchange and maybe storage I don't know I won't go into that right now but a medium of exchange of risk without having to go through the monetary Loop very very interesting yep I I can kind of imagine some areas Stephen and this this might be where bringing the body back in can help too as we you know you talk about like there's there's money in my hand and there's there's money like we talk about when you get it for something and then there's Capital which is almost like an augmented um cognition process which is up there in the markets you know that we've offloaded onto and is very hard to embody what's going on up there it's it's like it goes off it does its calculation it comes back and we do what I mean half the people doing it are doing on screens who themselves are being proxies for other people you know so there's there's there's a question about um where this comes back into ultimate um sense making at different time scales y yep you Ivan was smiling before I didn't were you smiling at the money thing I let's let's maybe have yeah let's have uh anything you want to go there Ivon if you want otherwise let's have a little closing imagin you and new new money and new money of new people yep so maybe we'll have a final set of thoughts so prepare that but the other side um Scott of the the money uh trading in the airport is that uh in a healthy system every sale is bidirectionally consensual every agreement is not feeling forced it's like so somebody wanted your peso the other person wanted the Yen so they made a deal they met in a Marketplace they wouldn't have found it on Craigslist the operating cost would have been too high but somebody wanted it now when does it become deranged well when there's um extra Market forces that force people to do things or when there's um you know devaluation Market manipulation etc etc but again it's not like there's a pure untrammeled Market out there and then it's like oh regulation how dare they it's actually we construct the market it's like we're building the bowl so it's not just like there's a pure bowl and then you've defiled it by putting Clay on it it's like no what is the structure that you're going to be interfacing in and can that not just be imbued with our values which aren't going to always have justification they're often acatic but can it actually allow diverse agents to act according to their deeply held axioms so that there is ability of people who have different faith in different things and it's been a joke for so long oh you don't need to believe in the legit the government to use the currency of a Fiat and now we're just seeing it in a new light so it's it's really interesting that market may be just an emergence from the fact of information seeking in trades yep and also leveraging and how that's trust like there's Bitcoin on the blockchain I just these are the examples I think about um yeah but then there's the 10x leverag Bitcoin who legitimized that exchange or that person to um offer 10x leverage or some other type of asset based upon leveraged on bitcoin nobody gave that permission if that person or that operation is reputable um and to whatever extent it is it will succeed or hopefully and again anything other than that is dysfunction so there's a way to talk about that in a way that is pretty different from the reward optimization of different competing agents that's previous Paradigm reward optimization competitive agents and we're not just moving from competitive to to Cooperative it's actually not that simple it's actually more about from the uh reward to the Precision guiding mindset and that everything else Falls in line after that and the action and the Precision are the key pieces some of the key pieces and so that's the coherence lending Factor yes we want to be precise about process together that's the minimal meme not we're g to have abundance in the future but today we're going to have Precision here yep so it's really yeah it's a it's a different it's really fascinating the way it's going to iterate itself into it's more scalable but it's not yet scaled y you know I I think of it makes me think of adaptation studies I have a book on Cambridge book on adaptation studies in theater where if you take Romeo and Juliet and you translate it into another language you got a lot of work to do because there's a lot of phrasing and there's a lot of behavior a lot of things for adaptation what we're doing here is is just like a giant simultaneous adaptation vehicle right because everyone's coming in with priors and this system allows people to discover things that they have in common it allows them to act in common and that's going to have such huge power it won't and it won't be something that will create spaces for it it will create the space this the interaction space will evolve and then be and then it'll be parameterized in metrics and you'll know if you're in it or not there'll be certain behaviors that indicate you're in it right but it's it's that's why I always say the process becomes the governance it's not you don't that's what a lot of people there's AI problem right now not R AI but the artificial intelligence that everyone is trying to create everyone that I can find is trying to create systems to govern the people creating the AI but not the AI system itself and I'm thinking more along lines of azimov which is you need the rules for the robots themselves similarly here each one of these systems is a robot it's not programmed by us per se but it's caused by us and it has uh features that are the result of our desires and needs so it's programmed in that way corporations are robots I think they're programmed our constructed Niche is how we we would consider it it you're never going to break away it's intertwin with us it's in our Niche yep go on right right and so the so it's interesting because I think a lot of folks look at Authority like the Fiat money is I think the U us is a Fiat not just because of rule of law but because of a monopoly of violence we have the most guns and bombs and so that it's a violent it's violence and imperialism and all that stuff that's wrapped up in Fiat right there's a power there and I think the power is shifting from physical violence to other forms of information I w't use the word violence here but information Authority and it feels like the ability of this system to declare that it can get you closer to reality continuously it's dynamically it's like the scientific method it's dynamically rejiggering itself I wouldn't say closer to truth I would say towards a better process together yes dynamically that's right yep that's right there's no there is no end state of Truth yep but that value proposition is not characteristic of the current systems and so it has a unique feature of D Dr risks in a way that's much uh more effective than the current systems it requires people to do the Cooperative thing which is like I said with credit cards or ATMs or swaps or anything with mitochondria right you know your free- existing things have to come together because it's a better deal to be had right so that's and so that so the convey our conveying why there's a better deal to be had we can use the existing problems in the world climate change War it's not even a better deal it's a better negotiating table yes and that's that's always the reframe is always from there will be better states to we're going to seek Precision today yes beautiful beautiful and that gets back to that now versus past and future thing that I was struggling with before right we're not going to we're not saying it's better it's a it's a better way to be and we don't know where being takes us because that's we have models of it you have a model I have a model we have a model let's get together figure out where we want to be the reality is the combin the synthesis of those desires I guess as evolving Through Time Clos closing statement from first Stephen then Ivonne if he want then back Scott well yeah I suppose that the the takeaway um and there was you had the E the G what was the third letter again there was um C is the preference Okay C I think that's really useful I think that's like a nice meeting point there between and I keep going back to this but ergodicity and noner it in the sense that you know as you get down to the smaller scales the body and I think even when you get into this Outsider cognition everything's got to follow that growth kind of erodic kind of churn and then you start to have this ability to bring in these non- erodic as we get to larger time scales and you know we even see as we get now to the state and how you can borrow money in ways which an individual can't we've seen this with the EU because they're not constrained by lifespan in the way that mortgage mortages are and so you've got this weird ability to bring in preferences in that um about of nonergodic sort of ways of just making a choice and creating deliberately steady state like a fe currency the whole point is it's steady and non erodic but we actually and this is where I think a lot the work with the indigenous cultures is they are they could say they're trapped in the old cyclical way of understanding the world but actually that gives you a way to tune into your deeper existential way of knowing about the long-term future because that is where your phenotype knowledge comes from not um you know the okay we have these structures that give us this ability to also do deep temporal depth but we do lose the ability to integrate free energy in such a new the dimensions come down and down and down and down to these kind of artifacts so yeah this is anyway this is what is coming up for me thanks well I hope Stephen that you and others can in the future you know listen back to what you've laid out and hopefully see it in a broader perspective what you were hinting at Ivonne any closing thoughts for 2020 indeed yeah sure I just want to thank you all for brilliant discussion and it's the great discussion for the of this uh strange Year and I hope to see you all next next from the beginning from the January and thanks for your help in the project from the beginning really critical yeah so very chill Scott closing thought you guys are this is this is beautiful you guys are beautiful this is awesome I'm I feel like a metamorphosis I'm I'm I'm I'm GNA pupate a little bit and then I'm going to come out after the New Year's and be ready to fly with you guys this is this is really delightful and fascinating and I think it's going to be really um very helpful to a lot of problems that are out there in the world it feels like there's a lot of um uh solution space that gets opened up with this and in in introducing it in in digestible chunks so I always talk to my clients about where are you now now where do you want to be that's a tough one and and now I see why it's a tough one because it's in a way the wrong question where do you where do we want to be individually and then how do we synthesize and then what are the ways to get there and then you triage them so that's a standard thing I used to use now that's changing because it's a um the exploration of where we want to be is the answer to that really hard second question which is people have difficulty answering so thank you for inviting me on the journey this is great stuff so to close um on that um Scott you just you laid out a helpful way of thinking about it with just where do you want to go this is common goal setting who are we where do we want to go how are we going to get there but you brought in two pieces which is the values the values are what is not explicitly described but they're part of the preference and the policy and so combining all of these disperate threads together and not having one Heat engine for the refrigerator and one power plant for the other one but combining all of these threads together so we can have a discussion from multi-perspective so that's the Intercultural component multi-perspective but also bringing together these coextn values that we have for system function exploration so many other features but really um my closing thought is for this live stream that uh tomorrow probably I'm going to record a video called 12.0 that is going to be the clock striking midnight on 2020 and then we will indeed head into a very exciting 2021 and Beyond so uh both of you uh well actually all of you I guess are uh RSD for the active inference lab so I'm just really looking forward to those three projects and just like you said Scott with the digestible bits live live stream has been our learning process we've all learned so much along this journey but now we all also want to be thinking about kic shorter content ones that have interactive Parts who knows but there's going to be yeah great things happening ahead so thanks everyone who's listening live or in replay thanks guys for participating fantastic and and happy New Year to everybody yeah happy year happy New Year wow what a discussion what a discussion all right well I'm going to to terminate the live stream so thanks everyone and I will talk to you later